$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathcal{S}}(t) = -\mathrm{i}\big[\hat{H}_{\mathcal{S}}, \hat{\rho}_{\mathcal{S}}(t)\big]$$
$$-\int_0^\infty \mathrm{d}\tau\, \left\{\nu(\tau)\left[\hat{\sigma}_z, \left[\hat{\sigma}_z(-\tau), \hat{\rho}_{\mathcal{S}}(t)\right]\right] - \mathrm{i}\eta(\tau)\left[\hat{\sigma}_z, \left\{\hat{\sigma}_z(-\tau), \hat{\rho}_{\mathcal{S}}(t)\right\}\right]\right\}. \quad (5.193)$$